UG (NEP) 5th SEMESTER
COMPUTER APPLICATIONS: DATA STRUCTURES USING C
(CAP522J2)

Session: December 2025 No. of Questions: 09
Time Allowed: 2:00 Hours Maximum Marks: 100 Minimum Marks: 36

NOTE: Attempt all questions from Sections “A” and “B” and only two questions from Section “C”.

SECTION A: Short Answer Type Questions (8 × 2 = 16 Marks)

Answer all questions in a maximum of 20 words each.

1. What are primitive data types?
2. What is a sparse matrix?
3. Define queue.
4. What is sorting?
5. Write the postfix expression of A+B-C.
6. Give the node structure of a singly linked list.
7. Mention the different ways of tree traversal.
8. List the graph traversal methods.

SECTION B: Medium Answer Type Questions (4 × 10 = 40 Marks)

Answer all four questions with internal choice in a maximum of 150 words each.

2. What is a data structure? Explain various types of data structures.

OR

Write an algorithm for binary search and discuss its speed compared with linear search.

3. Explain selection sort algorithm with suitable example.

OR

Convert the following infix expression into its equivalent postfix and prefix expressions: (a + b) (m/n) + (x + y).

4. Define linked list. Mention the applications of the linked list. Write a C program for searching a node in the singly linked list.

OR

Write a C program to perform insertion and deletion operations in a linked list.

5. Define Binary Search Tree. Give an example. Briefly explain various tree traversal methods with suitable examples.

OR

Define the terms (a) Adjacency Matrix (b) Adjacency List. Write a breadth-first-search algorithm.

SECTION C: Long Answer Type Questions (2 × 22 = 44 Marks)

Answer only two questions in a maximum of 400 words each.

6. What is a data structure? Distinguish between linear and nonlinear data structures. List the applications of data structures.
7. Given an infix expression: a+b*(c^d-e)^(f+g*h)-i. Convert it into postfix expression using stack.
8. What is a doubly linked list? Write a C program to insert and delete elements from a doubly linked list.
9. Construct an expression tree for the expression (a+b*c) + ((d*e+f)*g). Give the outputs when you apply inorder, preorder and postorder traversals.